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Wing spindle retention

Abstract

A wing conversion system is configured to rotate an engine of a vertical takeoff and landing aircraft relative to a wing section between flight modes. The wing conversion system includes a spindle, a conversion actuator, a drive ring, and a holding element. The spindle is rotatably supported for rotation with the engine and relative to the wing section about a spindle axis. The drive ring is drivingly engaged between the spindle and the output shaft, with the output shaft and spindle being rotatable with each other about the spindle axis. The holding element engages the drive ring and urges the drive ring into engagement with the output shaft and spindle.

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Background/Summary

BACKGROUND

1. Field

(1) Embodiments of the invention relate generally to aircraft structures with actuator devices. More specifically, the disclosed embodiments are related to wing conversion system associated with Vertical Takeoff and Landing (VTOL) aircraft.

2. Related Art

(2) It is known in the art for a tiltrotor aircraft to include an actuator mechanism for converting the aircraft between flight modes. For instance, U.S. Pat. No. 7,913,947 to Haynes et al. describes a system to pivot a pylon between two different flight modes. The system in Haynes has the actuator offset from the axis of rotation. U.S. Pat. No. 9,126,678 to Ross et al. describes the use of a linear actuator or rotary actuator, with a gear box coupling to a spindle. U.S. Pat. No. 9,663,225 to Kooiman et al. describes the use of a quill shaft with splined portions. U.S. Patent Application No. 2019/0277353 to Przybyla et al. describes a torque coupling mechanism with friction cones.

SUMMARY

(3) This summary is provided to introduce a selection of concepts in a simplified form that are further described below in the detailed description. This summary is not intended to identify key features or essential features of the claimed subject matter, nor is it intended to be used to limit the scope of the claimed subject matter. Other aspects and advantages of the invention will be apparent from the following detailed description of the embodiments and the accompanying drawing figures.

(4) In an embodiment, a wing conversion system is configured to rotate an engine of a vertical takeoff and landing aircraft relative to a wing section between flight modes. The wing conversion system broadly includes a spindle, a conversion actuator, a drive ring, and a holding element. The spindle is rotatably supported for rotation with the engine and relative to the wing section about a spindle axis. The conversion actuator is operable to rotate the spindle and includes a tubular output shaft, with the spindle axially extending at least partly through, and coaxial with, the output shaft. The drive ring is drivingly engaged between the spindle and the output shaft, with the output shaft and spindle being rotatable with each other about the spindle axis. The spindle and the drive ring include respective splines removably intermeshed with one another so that the spindle and drive ring rotate together. The output shaft and the drive ring include respective drive elements removably engaged with one another so that the output shaft and drive ring rotate together. The holding element engages the drive ring and urges the drive ring into engagement with the output shaft and spindle.

(5) In another embodiment, a method of assembling a wing conversion system configured to rotate

an engine of a vertical takeoff and landing aircraft relative to a wing section between flight modes is provided. The wing conversion system includes a spindle rotatably supported for rotation with the engine and a conversion actuator having a tubular output shaft. The method broadly includes the steps of extending the spindle axially through at least part of the output shaft; drivingly engaging a drive ring with the spindle and the output shaft so that the output shaft and spindle are rotatable with each other about the spindle axis; and engaging a holding element with the drive ring to urge the drive ring into engagement with the output shaft and spindle.

Description

BRIEF DESCRIPTION OF THE DRAWING FIGURES

- (1) Embodiments of the invention are described in detail below with reference to the attached drawing figures, wherein:
- (2) FIG. 1 is a fragmentary perspective view of a VTOL aircraft including a wing, engine, and an embodiment of a wing conversion system, with parts of the wing being removed to depict the wing conversion system supported relative to partitions of the wing;
- (3) FIG. 2A is a fragmentary perspective view of the wing conversion system shown in FIG. 1, depicting a conversion spindle, bearing blocks, and a conversion actuator;
- (4) FIG. 2B is a fragmentary perspective view of the wing conversion system similar to FIG. 2A, but showing a drive ring, compression ring, and retention nut of the wing conversion system exploded from the conversion spindle and the conversion actuator;
- (5) FIG. 3 is a fragmentary cross-sectional view of the wing conversion system shown in FIGS. 1-2B, depicting the drive ring engaged with an output shaft of the conversion actuator and a splined portion of the conversion spindle, with the drive ring secured by the compression ring and the retention nut;
- (6) FIG. 4 is an enlarged fragmentary cross-sectional view of the wing conversion system shown in FIGS. 1-3;
- (7) FIG. 5 is a fragmentary perspective view of the VTOL aircraft shown in FIG. 1;
- (8) FIG. 6 is a fragmentary cross-sectional view of the VTOL aircraft shown in FIGS. 1 and 6;
- (9) FIG. 7 is a fragmentary perspective view of the wing conversion system shown in FIGS. 1-6, depicting the drive ring engaged with the output shaft and the spindle and secured by the compression ring and the retention nut;
- (10) FIG. 8 is a fragmentary perspective view of the wing conversion system similar to FIG. 7, but showing the wing conversion system cross sectioned along a transverse section line;
- (11) FIG. 9 is a fragmentary perspective view of the wing conversion system similar to FIG. 7, but showing the wing conversion system cross sectioned along a transverse section line perpendicular to the section line of FIG. 8; and
- (12) FIG. 10 is a perspective view of a flange and a washer of the retention nut.
- (13) The drawing figures do not limit the invention to the specific embodiments disclosed and described herein. The drawings are not necessarily to scale, emphasis instead being placed upon clearly illustrating the principles of the invention.

DETAILED DESCRIPTION

- (14) The following detailed description references the accompanying drawings that illustrate specific embodiments in which the invention can be practiced. The embodiments are intended to describe aspects of the invention in sufficient detail to enable those skilled in the art to practice the invention. Other embodiments can be utilized, and changes can be made without departing from the scope of the invention. The following detailed description is, therefore, not to be taken in a limiting sense. The scope of the invention is defined only by the appended claims, along with the full scope of the equivalents to which such claims are entitled.

(15) In this description, references to “one embodiment,” “an embodiment,” or “embodiments” mean that the feature or features being referred to are included in at least one embodiment of the technology. Separate references to “one embodiment,” “an embodiment,” or “embodiments” in this description do not necessarily refer to the same embodiment and are also not mutually exclusive unless so stated and/or except as will be readily apparent to those skilled in the art from the description. For example, a feature, structure, act, etc. described in one embodiment may also be included in other embodiments, but is not necessarily included. Thus, the technology can include a variety of combinations and/or integrations of the embodiments described herein.

(16) Aircraft utilize a powered actuator as part of a wing conversion system. Upon liftoff of the aircraft, the wing and rotor may be converted from a vertical propulsion mode to a horizontal propulsion mode. Conversely, in preparation for landing of the aircraft, the wing and rotor may be converted from the horizontal propulsion mode to the vertical propulsion mode.

(17) Embodiments of the disclosure provide a wing conversion system, and a method of operating the system. FIG. 1 shows a perspective view of an embodiment showing incorporation of the disclosed embodiments into a vertical takeoff and landing (VTOL) aircraft **100**. Embodiments of the wing conversion system of VTOL aircraft **100** may provide a spindle powered by an actuator mechanism with high torque while minimizing both rotational and axial free play between the actuator and spindle.

(18) Turning to FIG. 1, embodiments of VTOL aircraft **100** may include a fuselage (not shown) mounted relative to a laterally extending wing **102**. One or more engines **104**, which may be used to drive one or more propellers (not shown), may be operably supported relative to the fuselage and wing **102** to provide aircraft propulsion. It is within the scope of the disclosure for embodiments of the aircraft to have an alternative fuselage and/or an alternative wing structure. The VTOL aircraft **100** may include the wing **102**, engine **104**, and a wing conversion system **106**.

(19) The depicted wing **102** may include an inboard wing section **110** and an outboard wing section **112** that extend laterally relative to one another and relative to the fuselage. Outboard wing section **112** may include a wing nacelle **114**, which operably receives a respective engine **104**. Wing sections **110** and **112** may include an outer skin **116** supported by partitions **118a** and **118b**. In the usual manner, partitions **118a** and **118b** are spaced laterally from one another to reinforce the wing structure.

(20) The VTOL aircraft **100** may be configured for conversion between a horizontal flight mode (see FIG. 1), where the engine is in a horizontal orientation, and a vertical flight mode (not shown), where the engine is in a vertical orientation. The wing sections **110** and **112** may be shiftably attached relative to one another to facilitate shifting of the engine **104** relative to the inboard wing section **110** between the horizontal flight mode and the vertical flight mode. As will be described, the wing conversion system **106** may pivotally interconnect the wing sections **110** and **112** and may permit common rotation of the outboard wing section **112**, the engine **104**, and the propeller with each other about a lateral spindle axis A and relative to the inboard wing section **110**.

(21) Referring to FIGS. 1-6, the wing conversion system **106** may comprise a powered mechanism for rotating the outboard wing section **112**. Embodiments of the wing conversion system **106** may broadly include a conversion spindle **130**, an inboard bearing block **132**, an outboard bearing block **134**, and a conversion actuator **136**. As will be described, the conversion spindle **130** may be fixed relative to the engine **104** and the outboard wing section **112** for rotation therewith about the lateral spindle axis A. The conversion actuator **136** may be configured to rotate the conversion spindle **130**, along with the engine **104** and the outboard wing section **112**, about the lateral spindle axis A for shifting the wing **102**, engine **104**, and propeller between the horizontal propulsion mode and the vertical propulsion mode.

(22) The depicted conversion actuator **136** may be drivingly attached relative to the conversion spindle **130** and operable to rotate the conversion spindle **130** between positions associated with the vertical propulsion mode and the horizontal propulsion mode. Conversion actuator **136** may

include an actuator housing **138**, a motor **140**, a transmission (not shown), and a tubular output shaft **142** powered by the motor **140** (see FIG. 2A). The actuator housing **138** of conversion actuator **136** may be fastened relative to the corresponding partition **118b**.

(23) The conversion spindle **130** may include an elongated tubular structure with an inboard section **130a** and an outboard section **130b**. Conversion spindle **130** may define a shoulder **144** located between the inboard section **130a** and outboard section **130b** (see FIG. 3). Shoulder **144** may abut an outboard end of the tubular output shaft **142**. Inboard section **130a** may include a splined portion **146** and a threaded portion **148** extending from the splined portion **146** to an inboard end **150** (see FIGS. 2B and 4). As will be described, the inboard section **130a** of conversion spindle **130** may be removably, drivingly attached to the output shaft **142** when assembling the wing conversion system **106**.

(24) Conversion spindle **130** may be rotatably supported relative to the inboard wing section **110** by inboard bearing block **132** and outboard bearing block **134** (see FIGS. 1, 5, and 6). Inboard bearing block **132** may include a bearing housing **152** associated with partition **118b** and an inboard roller bearing **154** operably supported within the bearing housing **152** (see FIGS. 3 and 6). Similarly, outboard bearing block **134** may include a bearing housing **156** associated with partition **118a** and an outboard roller bearing **158** operably supported within the bearing housing **152** (see FIG. 6). Thus, the inboard bearing block **132** and outboard bearing block **134** may be mounted relative to the inboard wing section **110** and spaced along an axial length of the spindle **130**. The conversion actuator **136** may be mounted adjacent the inboard bearing block **132** and configured to rotatably receive the spindle **130**.

(25) The inboard roller bearing **154** and outboard roller bearing **158** may rotatably support the conversion spindle **130** and permit spindle rotation relative to the inboard wing section **110** about the spindle axis A. In the depicted embodiment, the inboard roller bearing **154** and outboard roller bearing **158** are mounted on the outboard section **130b** of the conversion spindle **130**. The inboard section **130a** may extend axially at least partly through the output shaft **142** of the conversion actuator **136**. Preferably, when assembling the wing conversion system **106**, the inboard section **130a** of spindle **130** extends entirely through and is generally coaxial with the output shaft **142**.

(26) An inboard drive section **160** of the output shaft **142** may be drivingly attached to the inboard section **130a** by a split drive ring **162**. The split drive ring **162** may include drive ring halves **164** and **166** (see FIGS. 2A and 2B). The drive ring halves **164** and **166** may comprise arcuate drive ring sections that are identical to one another and arranged circumferentially. Each drive ring half **164** and **166** may have a curved body with an inboard portion **168** to engage the conversion spindle **130** and an outboard portion **170** to engage the output shaft **142** (see FIG. 4). In the illustrated embodiment, the inboard portion **168** may present inner splines **172** that receive and drivingly engage outer splines **174** of splined portion **146** of inboard section **130a** (see FIG. 4). Thus, the splines **172** and **174** may be removably intermeshed with one another so that the spindle **130** and drive ring **162** rotate together. Inboard portion **168** of each drive ring half **164** and **166** may also present an outer tapered surface **176** (see FIG. 4).

(27) The depicted outboard portion **170** may have circumferentially spaced drive dogs **178** (see FIGS. 2A and 2B). The drive dogs **178** of drive ring halves **164** and **166** may be removably intercalated with drive dogs **180** of the output shaft **142** (see FIGS. 2A and 2B) so that the drive ring **162** is removably engaged with the output shaft **142**. That is, the drive dogs **178** and **180** comprise drive elements that are removably engaged with one another so that the output shaft **142** and drive ring **162** rotate together. Although the illustrated drive dogs **178** and drive dogs **180** are preferred for providing removable driving engagement between the drive ring **162** and output shaft **142**, embodiments of the drive ring and/or the output shaft may have alternative drive elements (such as splines) for providing removable driving engagement between the drive ring and output shaft.

(28) When assembling the wing conversion system **106**, the drive ring **162** may be removably

drivingly engaged with the spindle **130** and the output shaft **142** so that the output shaft **142** and spindle **130** are rotatable with each other about the spindle axis A. For instance, the drive ring **162** may first be positioned on the spindle **130** so that splines **172** and **174** may be removably intermeshed with one another. This may be accomplished by moving the separated drive ring halves **164** and **166** radially toward each other until the drive ring halves **164** and **166** both engage the splines **174** of spindle **130**. Alternatively, the drive ring halves **164** and **166** may first be brought together (e.g., by partly inserting the drive ring halves **164** and **166** into conical compression ring **182**) and then moved axially onto the splines **174** of spindle **130**. With the drive ring **162** located on the spindle **130**, the drive ring **162** may be shifted axially to removably engage drive elements of the drive ring **162** and output shaft **142** with each other so that the drive ring **162** and output shaft **142** rotate together.

(29) While the drive ring preferably includes a pair of drive ring sections, it is also within the scope of various embodiments for the drive ring to include an alternative number of drive ring sections. For instance, various drive ring embodiments may include three or more drive ring sections arranged circumferentially. In at least certain embodiments, the drive ring may comprise a unitary ring structure.

(30) The wing conversion system **106** may further include the conical compression ring **182** and a retention nut **184** that cooperatively secure the split drive ring **162** in engagement with the output shaft **142** and the conversion spindle **130**. Compression ring **182** may comprise an endless holding element that engages the drive ring **162** to urge the drive ring **162** into engagement with the output shaft **142** and spindle **130**. Compression ring **182** may include an endless compression ring structure that presents an inner tapered surface **186** (see FIGS. 3 and 4). The compression ring **182** may be sized to at least partly receive the split drive ring **162**. Furthermore, the inner tapered surface **186** and the outer tapered surface **176** of the split drive ring **162** may be complementally shaped so that the compression ring **182** urges the split drive ring **162** into engagement with the conversion spindle **130** and the output shaft **142** (see FIGS. 3 and 4). In the depicted embodiment, the outer tapered surface **176** of the split drive ring **162** and the inner tapered surface **186** of the compression ring **182** may both comprise conical surfaces. However, embodiments of the outer tapered surface and/or the inner tapered surface may have alternative tapered surfaces that are complementary (such as planar surface shapes).

(31) When assembling the wing conversion system **106**, compression ring **182** may provide a holding element that urges the drive ring **162** into engagement with the output shaft **142** and spindle **130**. As the compression ring **182** is shifted axially (in an outboard direction) into further engagement with the split drive ring **162**, the radial force applied by the compression ring **182** onto the split drive ring **162** is increased (see FIG. 4). It will be appreciated that the compression ring **182** may apply a radial force that urges the ring sections (such as drive ring halves **164** and **166**) toward one another and into engagement with the spindle **130**. Again, axial movement of the compression ring **182** relative to the split drive ring **162** may also urge the drive ring **162** axially into engagement with the output shaft **142**.

(32) While the compression ring **182** may provide a preferred holding element to position the drive ring **162** in engagement with the output shaft **142** and spindle **130**, embodiments may include an alternative holding element. For example, embodiments of a holding element may include a plurality of discrete elements (such as elements spaced circumferentially along the drive ring) supported relative to one another. In various embodiments, a plurality of discrete elements may be supported relative to one another to cooperatively engage the drive ring. It will also be appreciated that one or more discrete elements may present tapered surfaces. Similar to the inner tapered surface **186**, tapered surfaces of one or more discrete elements may engage the outer tapered surface of the drive ring.

(33) The retention nut **184** may be used to apply a retention force to the compression ring **182** for securing the split drive ring **162** in engagement with the conversion spindle **130** and the output

shaft **142**. Retention nut **184** may include a flange **188**, washer **190**, fasteners **192**, and locking tabs **194**.

(34) Turning to FIGS. 7-10, flange **188** may define a threaded bore **208** (see FIG. 10) that permits the flange **188** to be selectively threaded onto the threaded portion **148** of the conversion spindle **130** (see FIGS. 4 and 7). Flange **188** may also present radial slots **210** (see FIG. 10). Washer **190** may define a smooth bore **212** (see FIG. 10) that is not threaded so that the washer **190** may be freely slidable on the threaded portion **148** without being threadedly engaged with the threaded portion **148**. Fasteners **192** may comprise set screws that are threaded into respective openings in the flange **188** and abuttingly engaged with the washer **190** (see FIGS. 7 and 8).

(35) The fasteners **192** may be axially shiftable relative to the flange **188** by threading the fasteners **192** into the flange **188**. Fasteners **192** may be axially shifted to move the washer **190** axially relative to the drive ring **162** (such as outboard movement of the washer **190** toward the output shaft **142**). As a result, fasteners **192** and washer **190** may be axially shifted to move a holding element (such as the compression ring **182**) axially relative to the drive ring **162** (such as outboard movement of the compression ring **182** toward the output shaft **142**).

(36) Locking tabs **194** of the retention nut **184** may be removably secured to the flange **188** with fasteners **214** and project radially inwardly from the flange **188** for engagement with slots **216** presented by the conversion spindle **130** (see FIGS. 7-9). The locking tabs **194** and conversion spindle **130** may be engaged with each other to restrict rotational threading movement of the retention nut **184** relative to the conversion spindle **130**.

(37) The threads of flange **188** may permit axial positioning of the retention nut **184** along the conversion spindle **130**. Thus, flange **188** permits retention nut **184** to be threaded into a position relative to the compression ring **182** (such as a position adjacent to or in engagement with the compression ring **182**). The fasteners **192** may be used to shift the washer **190** axially relative to the flange **188**. For example, with the retention nut **184** held onto the conversion spindle **130** by the locking tabs **194**, fasteners **192** may be threaded toward the washer **190** to advance the washer **190** into engagement with the compression ring **182**. Similarly, fasteners **192** may advance the washer **190** so that the washer **190** applies a gradually increasing force against the compression ring **182**.

(38) Although the invention has been described with reference to the embodiments illustrated in the attached drawing figures, it is noted that equivalents may be employed and substitutions made herein without departing from the scope of the invention as recited in the claims.

Claims

1. A wing conversion system configured to rotate an engine of a vertical takeoff and landing aircraft relative to a wing section between flight modes, said wing conversion system comprising: a spindle rotatably supported for rotation with the engine and relative to the wing section about a spindle axis; a conversion actuator operable to rotate the spindle and including a tubular output shaft, with the spindle axially extending at least partly through, and coaxial with, the output shaft; a drive ring being drivingly engaged between the spindle and the output shaft, with the output shaft and spindle being rotatable with each other about the spindle axis, said spindle and said drive ring including respective splines removably intermeshed with one another so that the spindle and drive ring rotate together, said output shaft and said drive ring including respective drive elements removably engaged with one another so that the output shaft and drive ring rotate together; and a holding element engaging the drive ring and urging the drive ring into engagement with the output shaft and spindle.

2. The wing conversion system as claimed in claim 1, said drive ring and holding element presenting respective tapered surfaces that are complementally shaped and slidably engaged with each other, with axial movement of the holding element relative to the drive ring operable to urge the drive ring axially into engagement with the output shaft and radially into engagement with the

spindle.

3. The wing conversion system as claimed in claim 2, said holding element comprising an endless compression ring.
 4. The wing conversion system as claimed in claim 2, said tapered surfaces being conically shaped.
 5. The wing conversion system as claimed in claim 4, said holding element comprising an endless compression ring.
 6. The wing conversion system as claimed in claim 1, said drive ring including a plurality of arcuate ring sections arranged circumferentially.
 7. The wing conversion system as claimed in claim 6, said holding element comprising an endless compression ring engaging and urging the ring sections toward one another.
 8. The wing conversion system as claimed in claim 7, said drive ring and holding element presenting respective conical surfaces that are complementally shaped and slidably engaged with each other, with axial movement of the holding element relative to the drive ring operable to urge the drive ring axially into engagement with the output shaft and radially into engagement with the spindle.
 9. The wing conversion system as claimed in claim 1, said drive elements of the output shaft and the drive ring each including a set of axially extending drive dogs, with the sets of drive dogs being intercalated with one another to restrict relative movement between the output shaft and the drive ring.
 10. The wing conversion system as claimed in claim 9, said drive dogs of each set being tapered toward a dog end, with the sets of drive dogs restricting relative rotational and axial movement between the output shaft and the drive ring.
 11. The wing conversion system as claimed in claim 1, said spindle having a threaded portion, and a retention nut threaded onto the threaded portion to position the holding element relative to the drive ring.
 12. The wing conversion system as claimed in claim 11, said retention nut including an endless body and fasteners shiftably supported by the body, said fasteners being axially shiftable to move the holding element axially relative to the drive ring.
 13. The wing conversion system as claimed in claim 11, said retention nut including a locking tab that removably engages the spindle to restrict threading of the retention nut relative to the spindle.
 14. The wing conversion system as claimed in claim 13, said retention nut including an endless body that presents a first tab slot to receive the locking tab, said spindle presenting a second tab slot, with the locking tab being removably secured in both tab slots to restrict threading of the retention nut.
 15. The wing conversion system as claimed in claim 1, bearing blocks configured to be mounted relative to the wing section and spaced along a length of the spindle, said spindle being rotatably supported by the bearing blocks.
 16. The wing conversion system as claimed in claim 15, said conversion actuator being mounted adjacent one of the bearing blocks.
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